

Jonathan P. Hyun

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EDUCATION

Duke University: B.S. Computer Science, B.S. Mathematics

August 2023 – May 2027

- Coursework: Data Structures & Algorithm, Design & Analysis of Algorithms, Database Systems, Machine Learning, High-Dimensional Data Analysis, Human-AI Teamin, Multi-Agent Systems **GPA: 3.90**

EXPERIENCE

MIT Lincoln Laboratory ([Website](#))

Lexington, MA

Software Engineering Intern, AI/ML Engineering Intern

April 2026 – Current

- Pioneered the first DDS (Data Distribution Service) tracking component in **C++**, distributing **1000 mbps** throughput computation across **52 decentralized servers**. Refined IDLs for efficient and standardized use.
- Innovated novel Convolutional Neural Network for OTHR improving baseline detection accuracy by **27%**.

Pulse LLC ([Website](#))

Durham, NC

Co-founder, Full Stack Developer

May 2024 – January 2026

- Engineered **full-stack** (React + Google Cloud + Firebase) web application to upload and distribute video feedback.
- Partnered with the Duke University School of Medicine for **active use in cohorts with 500+ MD students**.
- Implemented RAG framework on ChromaDB vector databases to generate LLM-tailored AI feedback.

Duke General Robotics Laboratory ([Website](#))

Durham, NC

AI Researcher, First Author, Undergraduate Mentor

February 2024 – Current

- Designed distributed training on lab servers with Docker Containers, scaling data collection for **60+ researchers**.
- Developed DOJO using C++, PyTorch, and CUDA, accelerating human-in-the-loop reinforcement learning for **50+ unique robots** with **10 state-of-the-art algorithms**, incl. PPO, DQN, DDPG, SAC, GUIDE, COACH.
- Presented work to 6 groups incl. Columbia's LIINC Lab, GWU's Mitroff's Lab, ARL DEVCOM, MIT Lincoln Lab.

Talk Hiring ([Website](#))

Remote

Software Engineering Intern, Front-End Developer

January 2023 – August 2023

- Developed Next.js interface to direct **20,400+ users** between Mock-Interview and Hire-Rewards programs.

PUBLICATIONS

CREW-WILDFIRE: Benchmarking Agentic Multi-Agent Collaborations at Scale | [Paper](#) | [Website](#) | [Video](#) | [GitHub](#) 2025

Jonathan Hyun, Nicholas R. Waytowich, & Boyuan Chen

Duke General Robotics Lab, US. Army Research Lab

- Accepted into **TMLR 2025** (Transactions on Machine Learning Research).
- Developed an end-to-end framework for hosting scalable large-scale Agentic teams in complex environments.
- Optimized distributed server/client Docker Containers with gRPC protocols to support up to **2,000+ AI Agents**.
- Benchmarked 5 state-of-the-art scalable multi-LLM Agentic algorithms across high-level behaviors.

PROJECTS

WILD-Teaming | React, WebSocket(TCP), AWS EC2, Docker, Python

- Designed and deployed a distributed full-stack multiplayer Human-AI teaming interface through Dockerized microservices on AWS EC2, supporting real-time, low-latency communication across nodes via WebSockets.
- Introduced a novel modular multimodal LLM/VLM-Agentic architecture, parallelizing communication, **reducing inference time by log(N)** across heterogeneous and dynamically scalable teams.

MiniAmazon | PostgreSQL, MongoDB, Flask, Python

- Architected full-stack marketplace through **CI/CD** pipeline with **20+ Flask APIs**, normalized database (10+ tables handling **authentication**, transactions, and inventory for **1000+ products** across multiple sellers.
- Optimized SQL queries (indexing, joins) to support **100+ concurrent transactions** and reduce order latency.

Languages: Python, Java, JavaScript/TypeScript, C, C++, SQL, Swift, CUDA

Frameworks / Libraries: React, RESTful API, Unity, Firebase, Next.js, Flask, PyTorch, gRPC, ROS

Developer Tools: Git, AWS (EC2, S3, RDS, DynamoDB, Lambda), Docker, Eclipse, MongoDB, Claude Code, Codex, Cursor